Project Title: Phishing Awareness Simulation Using Social Engineering Techniques

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1. Introduction

Cybersecurity is the practice of protecting systems, networks, and data from digital attacks. In today's digital world, cyber threats such as phishing, malware, ransomware, and data breaches are becoming increasingly common. According to cybersecurity studies, phishing is one of the most frequently used attack techniques by cybercriminals.

Phishing attacks involve sending fraudulent emails or creating fake websites to trick users into revealing sensitive information such as passwords, banking details, or personal data. These attacks exploit human psychology rather than technical vulnerabilities.

This project focuses on creating a **phishing awareness simulation** to understand how users react to suspicious emails and to educate them about cybersecurity threats.

2. Objectives of the Project

The main objectives of this project are:

• To understand phishing attacks and social engineering techniques.
• To simulate a phishing email and fake login webpage.
• To observe user behaviour when receiving suspicious emails.
• To analyze the results and identify user awareness levels.
• To suggest preventive measures to avoid phishing attacks.

3. Problem Statement

Phishing attacks are one of the most common cyber threats today. Attackers exploit human psychology and lack of awareness to gain unauthorized access to sensitive information.

Many users are unable to identify phishing emails or fake websites, which leads to security breaches and financial losses.

Therefore, it is important to understand how phishing attacks work and to educate users about identifying and preventing such attacks.

This project aims to simulate a phishing scenario in a controlled environment to analyze user responses and improve cybersecurity awareness**.**

4.Methodology

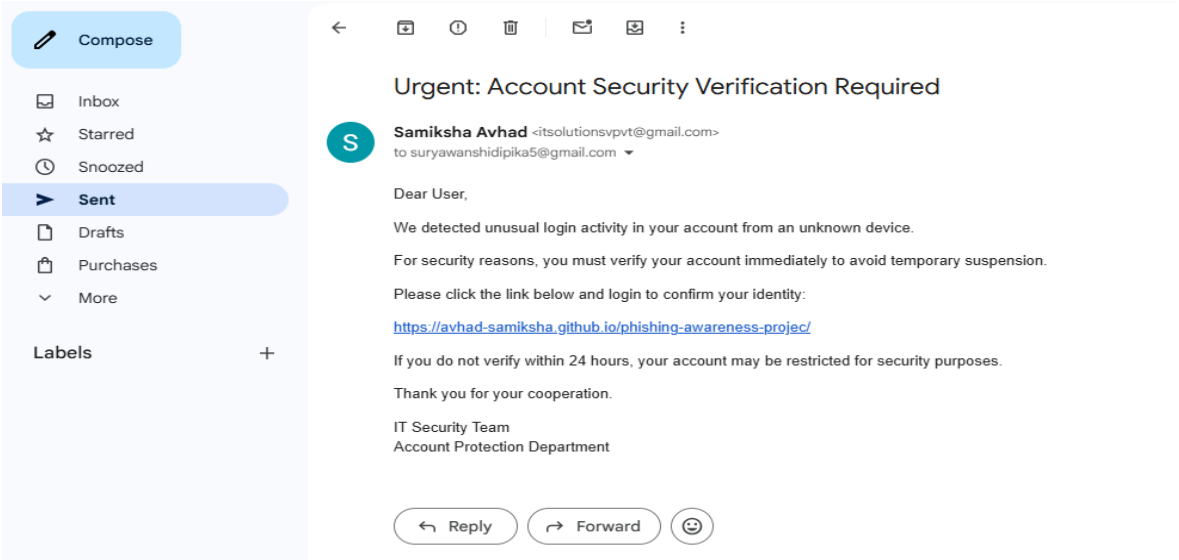
The project was conducted using a simulation approach. The following steps were performed:

1. Research was conducted on phishing techniques and cybersecurity threats.
2. A fake phishing email template was designed.
3. A fake login webpage was created using HTML, CSS, and JavaScript.
4. The phishing email was sent to a small group of participants.
5. Participants clicked the link and interacted with the simulated login page.
6. A Google Form survey was shared to collect user responses.
7. The collected responses were analyzed to understand user awareness.

This methodology allowed the study of user behaviour in a controlled and ethical environment.

5. Implementation

5.1 Phishing Email Simulation



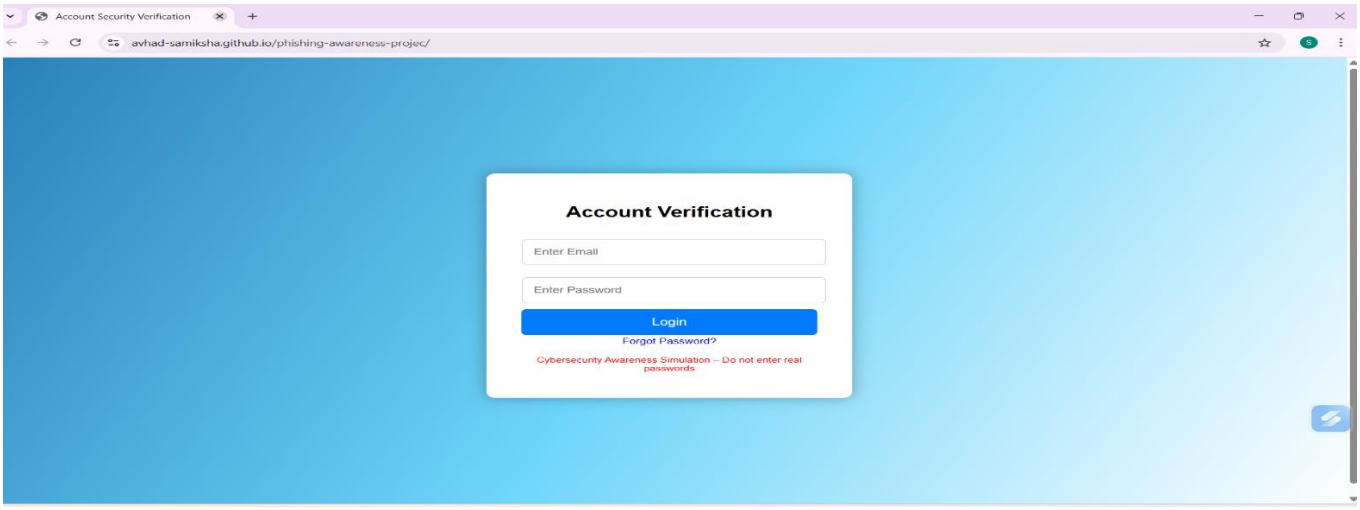
A phishing email was created that mimicked a security verification message. The email requested users to verify their account by clicking a login link.

The email contained:

- Subject line indicating urgent account verification
- Security warning message
- Link to the simulated login page

This technique replicates common phishing strategies used by attackers.

5.2 Fake Login Page



A login page was created using **HTML, CSS, and JavaScript**.

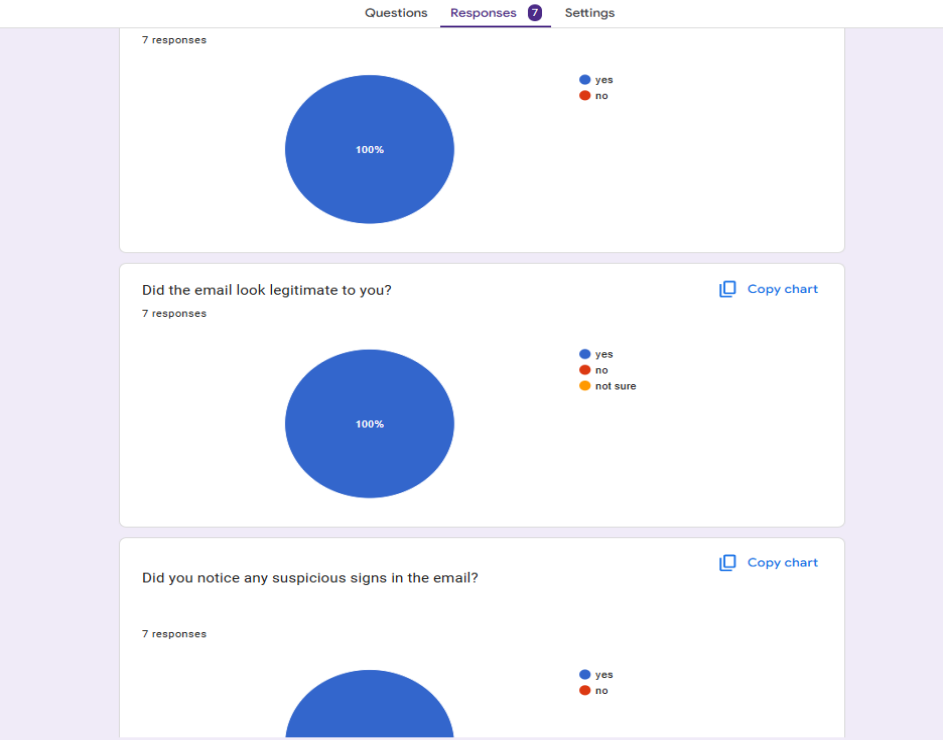
The page contained:

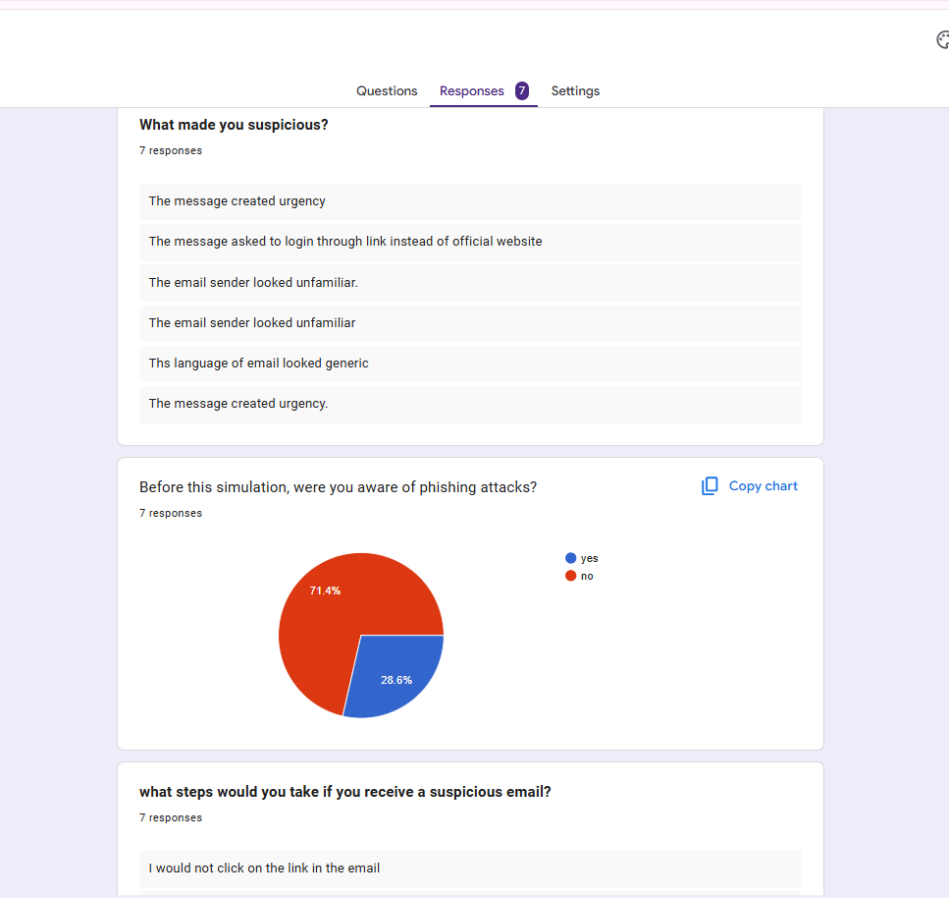
- Email input field
- Password input field
- Login button
- Error message validation

The page was uploaded to **GitHub Pages** to generate a public link.

This allowed participants to access the page via the phishing email link.

5.3 Google Form Survey





A Google Form survey was created to collect user feedback after the simulation.

Questions included:

- Did you receive the phishing email?
- Did the email appear legitimate?
- Did you notice anything suspicious?
- What steps would you take if you receive a suspicious email?

The responses were automatically collected and displayed in graphical format.

6. Results and Analysis

The simulation was conducted with a small group of participants.

Example results:

Total Participants	Clicked Link	Ignored Email
7	7	0

Analysis:

- Some users immediately clicked the link without verifying the sender.
- A few users suspected the email and ignored it.
- Some users realized the email was suspicious due to urgent language.

These results show that many users are still vulnerable to phishing attacks.

7. Lessons Learned

During this project, several important lessons were learned:

- Phishing attacks rely heavily on human psychology rather than technical weaknesses.
- Many users cannot easily identify fake emails.
- Urgent messages increase the likelihood that users will click malicious links.
- Awareness and education are essential to reduce cybersecurity risks.
- Organizations must implement security measures such as multi-factor authentication and employee training.

This project demonstrated how simple phishing techniques can successfully trick users if they are not careful.

8. Preventive Measures

To protect against phishing attacks, the following measures are recommended:

- Verify the sender's email address before clicking links
- Avoid clicking suspicious links in emails
- Use multi-factor authentication (MFA)
- Enable spam filters and email security systems
- Conduct cybersecurity awareness training
- Use antivirus and firewall protection

These practices significantly reduce the risk of phishing attacks.

9. Conclusion

Phishing is one of the most common cyber threats in modern digital environments. This project successfully simulated a phishing scenario to study user behavior and awareness.

The results demonstrate that many users are still susceptible to phishing attacks. Therefore, organizations and individuals must focus on cybersecurity awareness and preventive measures.

Educating users about phishing threats is essential to ensure safer digital environments.

10. References

• Cybersecurity Curriculum – Naviotech Solution
• OWASP Security Guidelines
• Cybersecurity Awareness Resources
• Google Forms Documentation